

AE-509: Dilution of [Ru(CO)(OH)2(PNP)] solution in MeCN while in-situ IR measurement

Date: 2025-07-03

Tags: Dihydroxo [Ru(CO)(OH)2(PNP)] AE

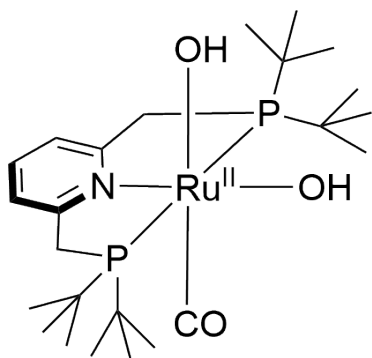
Future IR in-situ

Category: Two-photon

Status: Done

Created by: Alexander Eith

Reaction scheme/sample structure



Chemical Formula: $C_{24}H_{45}NO_3P_2Ru$

Molecular Weight: 558,64552

[AE-493.cdxml](#)

Literature/reference experiments

Literature	/
Reproduction	/
Similar experiments	Two-photon - AE-503: Dilution of [Ru(CO)(OH)2(PNP)] solution in water while in-situ IR measurement I

Reagents

Name	CAS Number / Experiment Number	Inventory number	Amount [μmol]	Equivalents	Mass _{theo} [mg]	Mass _{exp} [mg]	Molar mass [g/mol]	Volume [ml]	concentration [mM]
[Ru(CO)(OH)2(PNP)]	Two-photon - AE-456: Synthesis of [Ru(CO)(OH)2(PNPtBu)] using Ag2O, 5 h	/	5.44	/	/	3.04	558.65	/	27.2 --> 2.72 ^[a]
MeCN (dry)	AE-152: Degassing of MeCN	/	/	/	/	/	/	0.2 + 1.8	/
MeCN (HPLC grade)	75-05-08	/	/	/	/	/	/	1.0	/

[a]: obtained concentration range during experiment, stated mass was weighted into a flask

Procedure/observations

All steps were carried out using [\[Protocol\] Schlenk Technique](#), the flasks were not dried by heating prior to use, unless

other specified. All steps were carried out in the dark, unless other specified.

Date	Time	Step	Observations	Pictures
		The in-situ IR (Equipment - ReactIR 701L with DST 6.35 x 1.5 m x 305 DiComp probe) was used according to Protocol - In-situ IR measurment with ReactIR 701L in GL14 flask under inert conditions, the steps done are shortly explained in the experiment	/	/
		The following steps were done in air:	/	/
02.07	10:25	The device was started and cooled with liquid nitrogen	/	/
	13:55	To a GL14 5 mL flask (flask 1) MeCN (HPLC grade, 1 mL) was added using a 1 mL syringe	/	/
		The Probe was inserted into flask 1 and secured with the BOLA fitting	/	setup flask 1.jpg setup flask 1 full.jpg
		From now on stirring: 300 rpm for all samples and flasks during measurment		
	14:00	The background measurement was performed in MeCN (256 scans)	clear liquid	/
	14:02	The measurment of blank MeCN was started (sample interval 5 s, experiment duration: 8 h, resolution: 4 cm ⁻¹)	AE-509-1	/
	14:03	The measurment was stopped	/	/
	14:04	The probe was moved out of the liquid phase	/	setup flask 1 for -1-2.jpg
	14:04	The measurment of air was started (sample interval 5 s, experiment duration: 8 h, resolution: 4 cm ⁻¹)	AE-509-1-2	/
	14:11	The measurment was paused after 86 measurment points and the sample interval was changed to 3 min, afterwards the measurement was resumed	/	/
	14:17	The measurment was stopped	Total of 88 measurment points	/
		A new background measurement was performed in MeCN (256 scans)	/	/
	14:20	The measurment of MeCN was started (sample interval 3 min, experiment duration: 8 h, resolution: 4 cm ⁻¹)	AE-509-2	/
	15:40	To a GL14/NS14 5 mL flask (flask 2) [Ru] was added	/	/

		From here under inert conditons:	/	/
15:50	flask 2 was set under argon	greenish gray solid, partially stuck to the walls	solid.jpg	
15:55	The measurment was paused	/	/	
15:57	To flask 2 MeCN (dry, 0.2 mL) was added	dark orange brwon, looks like all dissolved	/	
15:58	The flask was swirled around for approx. 1 min	/	/	
15:59	The probe was switched to flask 2 in argon counterstream and subsequently the Glas stopper was switched with a new rubber septum	/	setup flask 2 0.2 mL.jpg	
16:00	The measurment of [Ru] in MeCN was resumed	/	/	
	New liquid N2 was added to the IR	This explains the spikes observed after 36 and 43 measurements, yielding a approx. 50 min wait, till measurment could be resumed	/	
17:10	In a 1 mL syringe MeCN (dry, 1 mL) was taken up without an argon buffer and quickly inserted	/	/	
17:12	MeCN (0.05 mL) was added from the syringe	56 measurments collected before addition	/	
17:24	MeCN (0.05 mL) was added from the syringe	60 measurments collected before addition	/	
17:33	MeCN (0.1 mL) was added from the syringe	63 measurments collected before addition	setup flask 2 0.4 mL.jpg	
17:42	MeCN (0.1 mL) was added from the syringe	66 measurments collected before addition	/	
17:51	MeCN (0.1 mL) was added from the syringe	69 measurments collected before addition	/	
18:00	MeCN (0.2 mL) was added from the syringe	72 measurments collected before addition	/	
18:09	MeCN (0.2 mL) was added from the syringe	75 measurments collected before addition	/	

	18:15	MeCN (0.2 mL) was added from the syringe	77 measurments collected before addition	/
	18:20	The now empty syringe was removed and a new 1 mL syringe filled with MeCN (dry, 0.8 mL) was added, again the MeCN was taken up without an argon buffer and quickly inserted	/	/
	18:21	MeCN (0.3 mL) was added from the syringe	79 measurments collected before addition during measurement 79 the glasfiber cable was slightly touched, yielding an offset in the data	/
	18:27	MeCN (0.5 mL) was added from the syringe	81 measurments collected before addition	/
	18:30	The measurment was stopped	82 measurments collected before addition	/

Analysis

Date	Time	Sample name	Analysis method	Analytical device	Solvent	Raw Data	Processed Data	Comparative Data	Interpretation
02.07	14:02	AE-509-1	in-situ IR	Mettler Toledo ReactIR 701L with Probe DST 6.35 x 1.5m x 305 DiComp	MeCN	AE-509-1.iCIR		/	blank MeCN, stopped after short time
02.07	14:04	AE-509-1-2	in-situ IR	Mettler Toledo ReactIR 701L with Probe DST 6.35 x 1.5m x 305 DiComp	MeCN	AE-509-1-2.iCIR		/	blank air, to get signals in spectrum for Moritz to test software
30.05	15:44	AE-509-2	in-situ IR	Mettler Toledo ReactIR 701L with Probe DST 6.35 x 1.5m x 305 DiComp	MeCN	AE-509-2.iCIR	AE-509-2.csv	Two-photon - AE-503: Dilution of [Ru(CO)(OH)2(PNP)] solution in water while in-situ IR measurement I Two-photon - AE-456: Synthesis of [Ru(CO)(OH)2(PNPtBu)] using Ag2O, 5 h (AE-456-IR-a)	first blank MeCN, than [Ru] in MeCN similar to spectra of solid

Data analysis

Step	Description	Used script	Interpretation	Obtained Plots/Data
1. Cleaning of data	Remove not usable data till measurement 49	in_situ_IR_cleaning_and_baseline.py	done, not really necessary, but kept anyway, since further processing needs it	AE-509-2_corrected_traces_original_49plus.csv AE-509_raw_and_cleaned.png
2. Concentration calculation	determine c in each dilution step	AE-509-conc.xlsx	Worked, range from 27 to 2 mM	ru_concentration_AE-509-2.csv
3. Conc plotting	Convert spectra from time domain to conc domain	in_situ_IR_concentration.py	Worked	AE-509-2_avg_per_condition.csv AE-509-2_conc.png
4. Baseline correction and Normilization	a. Baseline correction using assymetric least square fit for each spectra b. Normalize spectra to 1182 wavenumbers	in_situ_IR_baseline_correction_and_normilization_AE-503.py	Worked nicely, way less noise than in Two-photon - AE-503: Dilution of [Ru(CO)(OH)2(PNP)] solution in water while in-situ IR measurement I Clear signals observed in fingerprint area, but signals at 1600 and 1920 remain only useful data at 1600: with lower c: decrease at 1610 and increase at 1625 and 1635 at 1920: clear decrease a 1915 and weaker, but observable increase at 1945 and between 1960 and 1970	AE-509-2_normalized_corrected.csv AE-509-2_normalized_to_1182_zoom_1600-2.png AE-509-2_normalized_to_1182_zoom_1920-2.png AE-509-2_normalized_to_1182_all.png

Interpretation

Signals at 1600 and 1920 show pronounced change with changeing c --> emergence of two new peak each at higher wavenumbers with decreasing c --> could be sign for dimer(monomer equilibrium, but douplet typ signals increase with lower c, which I would rather assign the dimer than the monomer, espacially ifor the CO signal at 1920

Future recommendations

Old procedure	Problem	Suggested new procedure
Add IR probe to flask in Ar counterflow	some [Ru] sticked to the wall, so that they needed to be flushed down with MeCN before the IR probe could be attached	Be more carefull when adding the [Ru], use GL14 opening for addition, add IR Probe before addition of MeCN and than sekurate the flask with probe
Adding N2 (liquid) to the IR device during measurment	yields small shift in signal	do not do that, it´s could enough to measure several hours
IR probe touched after 79 measurments	yielded shift in signal	be careful not to touch the IR probe

Linked experiments

- [AE-152: Degassing of MeCN](#)

Prep work - [AE-442: Degassing of Milli-Q water](#)

Two-photon - [AE-456: Synthesis of \[Ru\(CO\)\(OH\)₂\(PNPtBu\)\] using Ag₂O, 5 h](#)

Two-photon - [AE-473: Testing of in-situ IR probe](#)

Two-photon - [AE-493: Dilution of \[Ru\(CO\)\(OH\)₂\(PNP\)\] solution in water while in-situ IR measurement](#)

Two-photon - [AE-499: Stocksolution of \[Ru\(CO\)\(OH\)₂\(PNP{tBu}\)\] in water](#)

Two-photon - [AE-503: Dilution of \[Ru\(CO\)\(OH\)₂\(PNP\)\] solution in water while in-situ IR measurement I](#)

Linked resources

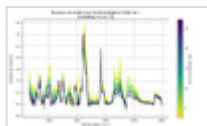
Equipment - [ReactIR 701L with DST 6.35 x 1.5 m x 305 DiComp probe](#)

Protocol - [In-situ IR measurment with ReactIR 701L in GL14 flask under inert conditions](#)

Attached files

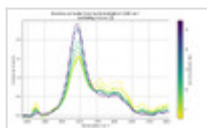
AE-509-2_normalized_to_1182_all.png

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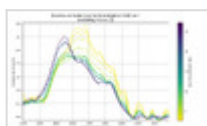
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AE-509-2_normalized_to_1182_zoom_1600-2.png

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AE-509-2_normalized_corrected.csv

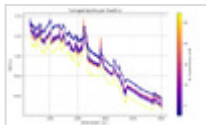
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in_situ_IR_baseline_correction_and_normilization_AE-503.py

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AE-509-2_conc.png

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AE-509-2_avg_per_condition.csv

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in_situ_IR_concentration.py

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AE-509-2_corrected_traces_original_49plus.csv

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ru_concentration_AE-509-2.csv

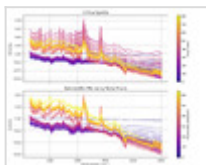
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AE-509-conc.xlsx

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AE-509_raw_and_cleaned.png

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in_situ_IR_cleaning_and_baseline.py

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setup-flask-2-0.4-mL.jpg

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setup-flask-2-0.2-mL.jpg

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solid.jpg

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setup-flask-1-for-1-2.jpg

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setup-flask-1-full.jpg

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setup-flask-1.jpg

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AE-509-2.iCIR

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AE-509-2.csv

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AE-509-1-2.iCIR

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Comment

On 2025-07-15 11:33:19 Jacob Schneidewind wrote:

Notes:

- * Reagent table: deleting [Ru 2]
- * Reagent table: specifying concentration range in concentration column
- * 14:00: specifying that solution was stirred during background measurement
- * Specifying what was measured in measurement entries, e.g. 14:02: measurement (blank MeCN) was started
- * 14:20: specifying that probe was moved back down into liquid phase
- * Future recommendation: assembling measurement flasks (NS14/GL14) before putting it under argon
- * Adding future recommendation: not refilling liquid N2 during measurement
- * 16:00: specifying that a 50 min wait was necessary for the signal to stabilise
- * Adding breakdown of measurements before measurement 56 (unusable measurement, useful measurements for initial concentration, useful measurement for blank
- * Adding future recommendation: being careful not to touch the fiber
- * Analysis table: in interpretation briefly specifying what was measured (blank MeCN, blank air etc.)



Unique eLabID: 20250703-b83d6dfa3b4b75395084f864a3022d0a9d476207
Link: <https://elab.water-splitting.org/experiments.php?mode=view&id=2386>